

RESEARCH ARTICLE

Real-Time Fire Detection From Video Surveillance Cameras: Where Are We Now?

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ABSTRACT Automatic fire detection in video surveillance is a critical capability for safeguarding industrial sites, public infrastructures, and natural environments. Although recent advances in deep learning have substantially improved early fire detection, the performance of existing methods remains highly dependent on scene conditions such as viewing range, background activity, and the presence of fire-like distractors. This paper presents a comprehensive evaluation of real-time fire detection techniques on the ONFIRE and FIRE-TASTIC benchmarks, currently the most extensive datasets available for video-based fire monitoring. Through a systematic comparison of methods that differ in detector quality, temporal modeling, and semantic reasoning capabilities, we highlight the strengths and limitations of current design paradigms and analyze their behaviour across scenarios that differ in terms of range and background activity. Our results reveal clear performance trends, emphasizing the crucial role of careful detector training, scenario-aware adaptation, temporal analysis, and lightweight vision–language integration. Building on these insights, we outline promising research directions, including the development of more representative datasets, improved semantic confirmation modules, and robust continual learning strategies capable of maintaining reliability in dynamic real-world deployments.

INDEX TERMS Fire detection, video surveillance, real-time computer vision, deep learning, temporal analysis, vision–language models, scenario-aware detection.

I. INTRODUCTION

Fire remains one of the most impactful hazards to human life, built environments, and natural ecosystems [1]. The risk is particularly elevated in industrial and rural zones where combustible materials, electrical infrastructures, and machinery are frequent sources of ignition. Moreover, the effects of climate change are amplifying this threat: the global rise in temperature [2] is increasingly associated with extended drought periods and severe weather conditions, which exacerbate the frequency and intensity of fires worldwide [3], [4]. In this context, the ability to recognize fire events at their earliest stages is essential for mitigating losses and supporting rapid emergency intervention.

In recent years, video surveillance systems have become widely deployed as part of environmental and critical

infrastructure monitoring strategies. This trend has accelerated the need for robust automatic fire detection methods based on computer vision. Early research predominantly relied on low-level color cues or simple motion descriptors extracted from camera streams. Although computationally lightweight, these methods cannot cope with the substantial variability in real-world video data, such as illumination changes, distance from the camera, atmospheric interference, and camera noise.

More advanced approaches grounded in machine learning and deep neural networks have significantly improved detection sensitivity. Nevertheless, they still face two persistent challenges. First, the visual appearance of fire can vary considerably depending on scale, viewpoint, and scene dynamics, which often differs from what is represented in the training data. This results in missed detections when flames or smoke are small or partially occluded. Second, non-fire objects exhibiting flame-like color or motion patterns may

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trigger false alarms, especially in the presence of vehicle lights, moving machinery, or clouds.

Recent analyses of the field highlight two major gaps. The first concerns the lack of scenario awareness in most fire detection pipelines [5]. Surveillance footage can involve scenes where the fire source is close to the camera and clearly visible (short-range), but also settings where flames occupy only a few pixels or appear intermittently due to occlusion or smoke dispersion (long-range). Moreover, the level of background activity, ranging from quiet static environments to highly dynamic scenes with animals, vehicles, machinery, or vegetation, has a direct impact on the type and frequency of fire-like distractors. The taxonomy introduced in [5] organizes these factors into four representative scenarios (low activity short range, low activity long range, high activity short range, high activity long range), underscoring how range and activity jointly define the difficulty of detection. Systems that fail to adjust their inference strategy to these contextual factors often exhibit inconsistent performance, showing high sensitivity in simple scenes but suffering from false alarms or missed detections in complex or long-range cases.

The second gap, confirmed by the ONFIRE contest results [6], [7], is that even recent detectors and more complex approaches, despite being highly sensitive, may still produce excessive false positives. This limitation often stems from training datasets dominated by simplistic or low-activity scenes, which reduce the ability of the methods to generalize to more realistic conditions where various moving or luminous objects mimic fire appearance. These shortcomings highlight a broader challenge in current research: achieving robust performance across the entire spectrum of real-world surveillance contexts, from easy short-range captures to difficult high-activity scenes rich in fire-like distractors.

Motivated by these observations, this work provides a systematic analysis of the current state of real-time video-based fire detection. Specifically, we evaluate existing approaches under the challenging and diverse conditions represented in the ONFIRE [8] and FIRE-TASTIC [9] test sets, currently the most extensive benchmarks for fire detection in surveillance scenarios. By reviewing how different detection strategies behave across varying scene complexities and scenario categories, this study analyzes and clarifies the limits of current methodologies and outlines emerging research directions aimed at bridging the gap between high laboratory performance and robust real-world operation. In particular, our study points out the importance of scenario-aware models, richer datasets, more effective temporal and semantic reasoning, and advanced adaptation mechanisms for improving the robustness of future fire detection systems.

II. RELATED WORKS

The use of conventional optical cameras for fire detection has been investigated since the early 2000s due to their

low cost and ease of deployment. Early vision-based systems primarily relied on heuristic color thresholding in tailored color spaces [10], [11], sometimes complemented by smoke-specific segmentation modules to capture early-stage cues [12]. While computationally lightweight, these handcrafted approaches were highly sensitive to illumination variations and easily confused fire-like motions, resulting in poor robustness in complex environments.

To overcome these limitations, temporal reasoning was progressively integrated through flicker analysis, motion stability checks, or shape evolution descriptors [13], [14], [15], [16], [17], [18], [19]. Although these multi-stage rule-based systems improved reliability, their manually designed features limited scalability across diverse scenarios.

The introduction of machine-learning techniques enabled richer texture and appearance modeling, such as Local Binary Patterns [20] combined with SVMs or ensemble methods [21], [22], [23]. These approaches outperformed purely handcrafted pipelines but remained constrained by the expressive limits of manually engineered descriptors.

With the advent of deep learning, fire detection shifted toward data-driven representations. One research direction focuses on single-image analysis: either image-level classification [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34] or direct bounding-box localization using efficient CNNs. Lightweight classifiers such as FireNet [32], KutralNet [26], and MSDNet [33] demonstrate that compact architectures can achieve fast inference on embedded devices, although they struggle when fire occupies only small regions or when fire-like distractors dominate the scene. In parallel, single-frame object detectors gained popularity thanks to their ability to explicitly localize flames or smoke. YOLO-based detectors fine-tuned on large annotated datasets, e.g., ONFIRE in [7] or the diverse FOCUS dataset [35], have shown strong performance even on small distant targets, confirming the central role of high-quality heterogeneous training data.

Given the limitations of single-frame reasoning, temporal analysis has re-emerged as a crucial component in modern pipelines. Recent systems integrate spatial detectors or classifiers with temporal verification schemes, ranging from simple persistence rules [36] to more sophisticated confidence aggregation strategies [6]. Deep sequence models now play a prominent role as well: 3D CNNs, two-stream architectures, recurrent networks, and Transformer-based encoders jointly model appearance and motion [37], [38], [39], [40], [41], [42], [43], [44]. These hybrid spatio-temporal methods can reduce false alarms by verifying the persistence, coherence, and characteristic dynamics of fire, although they demand larger and more diverse video datasets to avoid overfitting.

Most recently, a new family of approaches incorporates vision-language models (VLMs) to improve semantic discrimination [45]. Because VLMs are pre-trained on millions of image-text pairs, they excel at recognizing subtle visual

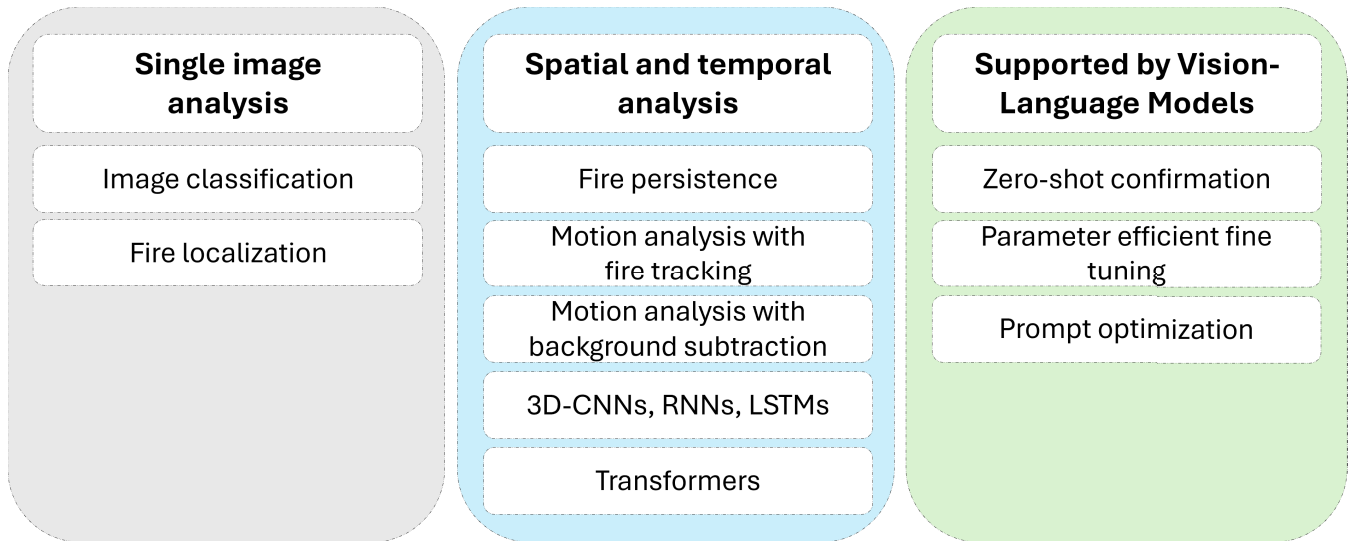


FIGURE 1. Taxonomy of recent real-time fire-detection methods, depicting the three major families of approaches discussed in this work. Single-image analysis methods rely solely on frame-level cues, typically using image classifiers or simple localization models without temporal reasoning. Spatial and temporal analysis approaches extend this paradigm by integrating mechanisms that track fire candidates over time, improving robustness to transient artifacts. Finally, methods supported by vision-language models combine spatial-temporal analysis with semantic confirmation, enabling a higher level of contextual understanding and improved discrimination between true fire events and visually similar distractors.

concepts and distinguishing genuine fire from visually similar objects. Systems such as FIRE-TASTIC [9] and FOCUS [35] first rely on YOLO-based detectors to identify frame-level fire candidates and apply temporal filtering, querying a model like BLIP-2 only when persistent regions require semantic confirmation. This selective validation strategy dramatically reduces false positives while preserving high sensitivity, especially in visually challenging or high-activity environments.

In conclusion, the evolution of fire detection reflects a progression from handcrafted color models to deep spatio-temporal architectures and, more recently, to multi-modal pipelines augmented with vision-language reasoning.

III. FIRE DETECTION METHODS

In the following sections, we outline the recent fire detection methods which published results on the benchmark datasets (that we will analyze in Section IV-A) and that exhibited the capability to run in real-time. We have divided them into the three main categories depicted in Figure 1. The first comprises approaches that operate on single images, either through image-level classification or direct spatial localization of flames and smoke. The second includes methods that integrate both spatial and temporal information, exploiting motion patterns and persistence across consecutive frames to improve reliability. The third category gathers approaches supported by vision-language models, which leverage large-scale image-text pre-training to semantically validate visual detections and better discriminate fire from fire-like objects. Table 1 summarizes the considered methods, categorized according to the proposed taxonomy.

A. SINGLE-IMAGE ANALYSIS

In this section we describe the methods focused on single-image analysis, relying either on image-level classification or direct detection of flame or smoke regions within individual frames. These methods operate without temporal cues, making them computationally efficient but inherently sensitive to fire-like visual ambiguities.

Fire detection methods based on image classification treat each frame as an independent image, assigning a fire/non-fire label without leveraging temporal cues. The method called FireNet [32] relies on a lightweight CNN classifier (FireNet) that distinguishes fire from non-fire content using single images. To ensure robustness, the authors constructed a training set by combining samples from existing datasets with newly collected and augmented images containing real fires and fire-like distractors, enabling the model to learn more generalizable visual cues. The resulting network, optimized for efficiency with only three convolutional layers and regularized through dropout, was trained end-to-end and was able to run in real-time on low-cost embedded hardware. KutralNet [26] is a compact deep learning architecture for fire detection designed to significantly reduce computational cost, requiring approximately 92% fewer floating-point operations than FireNet. It serves as a lightweight baseline for evaluating different efficient convolution strategies, including inverted residual blocks, depthwise convolutions, and octave convolutions. The network was trained and tested using two image datasets, FireNet and FiSmo, encompassing diverse fire and non-fire samples to ensure robust performance in realistic scenarios. The approach proposed by Int. Systems Lab [33] also relies on single-frame classification. It introduces a Multi-Scale Dense Network (MSDNet),

TABLE 1. Summary of the considered fire detection methods according to the proposed taxonomy. For each approach, the table reports: (i) the technique used for single-image analysis, (ii) the presence and nature of spatial and temporal analysis, and (iii) whether semantic confirmation with a vision-language model is employed.

Method	Single-image analysis	Spatial and temporal analysis	Support by vision-language models
FireNet [32]	Image classification	-	-
KutrolNet [26]	Image classification	-	-
Int. Systems Lab [33]	Image classification	-	-
ShuffleNetV2 [34]	Image classification	-	-
Tendeiro [7]	Fire localization	-	-
Yolov5s [35]	Fire localization	-	-
Yolov8s [35]	Fire localization	-	-
Yolov11s [35]	Fire localization	-	-
Yolov12s [35]	Fire localization	-	-
KIOS-EVAI [6]	Image classification	Fire persistence	-
FIRESTART [42]	Image classification	Fire persistence	-
MADFIRE [6]	Image classification	Fire persistence	-
Slow-Fast [6]	Image classification	3D-CNN	-
State Key Lab [6]	Image classification	Transformer	-
Iplab@Unica [43]	Image classification	Fire persistence	-
Cheng et al. [44]	Image classification	GRU	-
AI-KP [6]	Fire localization	Fire persistence	-
De Venancio et al. [36]	Fire localization	Fire persistence	-
FireBusters [6]	Fire localization	Fire persistence	-
FLAME [41]	Fire localization	Motion analysis and fire persistence	-
FIRE-TASTIC [9]	Fire localization	Motion analysis and fire persistence	Zero-shot classification
FOCUS [35]	Fire localization	Motion analysis and fire persistence	Zero-shot classification

which operates as a cascade of three image classifiers built upon MobileNet-v2, EfficientNet-B0, and ResNet-50. When an input image is straightforward, the first classifier issues a prediction as soon as its confidence exceeds a predefined threshold, minimizing computation; for more challenging cases, subsequent classifiers are invoked until a sufficiently confident decision is reached. This design enables adaptive allocation of computational effort based on image complexity. The model was trained solely on the ONFIRE training set, with no additional data augmentation or external samples. The method proposed in [34] introduces a lightweight CNN based on ShuffleNetV2, derived from a well-known architecture but substantially simplified to reduce computation while preserving accuracy. The model is used both for full-frame binary classification and for high level fire localization by applying SLIC superpixel segmentation, where each segmented region is independently classified as fire or non-fire. Because these methods operate solely at the image level without explicitly localizing the fire, they can struggle when flames or smoke occupy only a small portion of the frame, as in long-range conditions. Similarly, when visually similar distractors appear in dynamic scenes with high activity, these approaches are not enough reliable.

Methods based on object detectors offer the advantage of explicitly localizing fire and smoke regions in the image, including small or distant instances. However, they need fully annotated training datasets with precise bounding boxes for every occurrence, making data preparation significantly more demanding. Tendeiro's approach [7] employs YOLOv8n, a lightweight object detector chosen to balance speed and

accuracy, and fine-tunes it on the ONFIRE training set, after annotating flame and smoke instances with bounding boxes. The model is optimized using a joint loss for objectness, classification, and localization, with data augmentation applied to improve robustness across varying surveillance conditions. During inference, frames are sampled at 1 FPS and passed through the detector, which triggers an alarm once flame or smoke is identified above a confidence threshold. The YOLOv5s, YOLOv8s, YOLOv11s and YOLOv12s detectors described in [35] were all trained on the FOCUS dataset, whose careful collection and preparation represent the point of strength of these methods. Since the detector performance is inherently tied to the quality and diversity of the training data, the dataset is a large, heterogeneous, and well-balanced collection of flame, smoke, and challenging non-fire samples. It integrates frames extracted from the ONFIRE training set, a large portion of the D-Fire dataset, and more than thirty thousand additional images collected from the web and public datasets. Particular attention was given to ensuring diversity across distances, lighting conditions, weather variations, geographical contexts, and especially to including difficult negatives containing fire-like objects. This results in a dataset with a high percentage of negative samples and a balanced number of flame and smoke annotations, providing strong support for both sensitivity and robustness.

As we will show in Section V, these approaches achieve a limited F-score (less than 85%), mainly due to the limited precision in discarding false positives, that is around 75% in the best cases, but around 60% in the average.

B. SPATIAL AND TEMPORAL ANALYSIS

In this section, we present the approaches that integrate both spatial and temporal information to perform fire detection, leveraging sequence of frames to verify the persistence of flames or smoke and to capture the characteristic motion patterns of fire cues. These methods go beyond single-image analysis by incorporating temporal reasoning, which helps to reduce false alarms improving the precision in realistic surveillance scenarios. These methods can be distinguished with respect to the strategy used to detect fire in individual frames, ranging from image classification to fire localization, and by the specific techniques employed to assess temporal consistency and motion dynamics across consecutive frames.

Among the approaches using image classification, the KIOS-EVAI method [6] performs frame-level fire detection using a modified EfficientNet-B0 classifier, enhanced with a CBAM attention module to better focus on fire-relevant spatial and channel features. The model is trained on an extended version of the ONFIRE dataset, augmented with additional private forest-fire videos, which sampled at 1 FPS yield over 120,000 labeled frames. Temporal consistency was enforced through a sliding-window strategy: the system computes the average classification confidence over one second of video, and a fire alarm is issued only when this aggregated score exceeds a predefined threshold. FIRESTART [42] detects fire at the frame level using a MobileViTv2-based classifier that predicts whether each image belongs to a pre-ignition or post-ignition time range. After this initial classification, the method applies a temporal post-processing step that analyzes the evolution of confidence scores within a sliding window to refine the estimated ignition moment and reduce notification delay. The network is trained using the ONFIRE dataset enriched through data augmentation, including flips, rotations, and random crops. The MADFIRE approach [6] detects fire on a frame-by-frame basis using a modified ResNet-50 classifier, enhanced with nine additional fully connected layers, eight with ReLU activations and one sigmoid output, and multiple dropout layers to improve generalization. The model is trained on an extended dataset that supplements the ONFIRE dataset with additional fire imagery from other sources. To ensure temporal consistency, the system requires fire to be detected in 5 consecutive frames before issuing a final alert. The Slow-Fast approach [6] performs joint spatial and temporal fire detection using a 3D-CNN built on the Slow-Fast-R50 architecture. Binary classification is done on short video clips rather than individual frames: the slow pathway extracts high-level semantic cues at low frame rates, while the fast pathway captures rapid temporal variations characteristic of flames and smoke. The original classifier is replaced with a single-output sigmoid layer to perform binary video-level fire detection. To train the model, the ONFIRE dataset was extended through diverse augmentation techniques, such as Gaussian blur, horizontal flipping, and random adjustments in brightness, contrast, hue, and saturation, and by adding

negative samples from the web to mitigate class imbalance. The method proposed by State Key Lab [6] performs fire detection by jointly analyzing spatial and temporal cues through a Video Swin Transformer (Video SwinT). The approach processes short clips of consecutive frames, where shifted-window self-attention allows the model to capture flame or smoke patterns that persist or evolve across frames. After feature extraction, a lightweight classification head, featuring 3D max pooling, dropout, and a fully connected layer, produces the final decision while maintaining computational efficiency. For training, the authors used the ONFIRE dataset enriched with additional private videos containing positive fire examples, increasing the sensitivity of the proposed approach. The Iplab@Unica method [43] detects fire by performing temporal classification over sequences of 16 consecutive frames. Individual frames are first encoded using the OpenGVLabVideoMAE v2-Base model to extract video embeddings, which are then processed by a GPT-2-based classifier fine-tuned with LoRA for multi-label prediction of no event, flame, or smoke. Temporal consistency is enforced via a smoothing filter, which confirms a fire detection only if at least 3 consecutive sequences predict fire, helping to reduce false alarms and refine the estimated onset time of events. Cheng et al. proposed a method [44] which detects fire by first extracting spatial features from individual frames using a DenseNet convolutional backbone, and then models temporal dynamics across video clips with a GRU recurrent module. The model outputs a probability score for each clip, while temporal consistency is implicitly captured by the recurrent GRU, allowing the system to account for the persistence and motion of flames or smoke over consecutive frames.

As already discussed in the previous section, methods based solely on image classification tend to be less effective, particularly in complex scenarios. To address these limitations, a range of approaches has been developed that rely on explicit fire localization, training detectors on carefully curated datasets annotated with bounding boxes for flames and smoke to improve both sensitivity and robustness. The AI-KP approach [6] performs frame-level fire detection using a YOLOv8n one-stage detector, which identifies fire regions and outputs corresponding bounding boxes for every video frame. The model is trained on a manually annotated dataset of 67,536 images, collected from the ONFIRE dataset and expanded with data augmentation. To ensure temporal consistency and reduce false alarms, the system raises an alert only when fire is detected in at least N consecutive frames, thus verifying the persistence of the flame over time. The method proposed by De Venâncio et al. [36] detects fire in individual frames using a YOLOv5 object detector trained on the D-FIRE dataset. Temporal consistency is assessed through a rule-based analysis that verifies the persistence of detected fires across 30 consecutive frames, while also tracking the position and size of the corresponding bounding boxes to reduce false alarms and confirm genuine fire events. The

FireBusters method [6] performs frame-level fire localization using a single-stage YOLOv5s detector and verifies temporal consistency through a confidence-accumulation mechanism across consecutive frames. To build a reliable training set, the authors extracted one frame per second from every video in the ONFIRE dataset and manually inspected the resulting images, removing near-duplicate samples and retaining those that provided complementary visual information. Since the initial dataset was strongly imbalanced toward smoke, they expanded it by collecting additional flame-only, smoke-only, mixed flame-and-smoke, and challenging negative examples from the web, assembling 32,589 annotated images with bounding boxes for each fire instance. Temporal persistence was assessed by accumulating YOLOv5s detection confidences over time: if any pixel's cumulative value exceeded a configurable threshold within a window of 0.5 seconds, a fire alarm was triggered. FLAME [41] adopts a modular approach, combining frame-level fire localization with motion-based verification to achieve high recall and precision. In the first stage, a fire detector based on YOLOv8s localizes flames and smoke separately in individual frames, learning distinct appearance features while being trained on a carefully curated set of positive and challenging negative samples to reduce false positives. The second stage applies background subtraction to retain only moving candidates, filtering out stationary fire-like objects; moreover, it performs a model-based motion analysis using a tracking-by-detection algorithm to confirm the persistence and coherence of moving fire candidates over time before triggering an alarm.

We can conclude that the methods combining spatial and temporal analysis demonstrate increasingly robust performance as the precision of fire localization and the representativeness of the training data improve, and as the temporal verification mechanisms become more sophisticated. By evaluating the persistence and motion of fire cues over time, these approaches are better able to distinguish genuine fire events from transient noise, achieving higher recall and precision than single-frame detectors. It is worth mentioning that the success of these approaches strongly depends on the effectiveness of the fire detector, substantially more accurate than image classification, and on the temporal analysis. As shown in Section V, only the best methods achieve F-score higher than 85%, while the others obtain performance similar to the methods performing single-image analysis. Nonetheless, also the best methods in this category can still be challenged by persistent fire-like objects, which may occasionally be misclassified as fire as shown in Figure 2, highlighting the need for further refinement in spatial discrimination.

C. SUPPORTED BY VISION-LANGUAGE MODELS

In this section, we describe two recent methods that incorporate vision-language models to support or confirm fire detection. The rationale behind these approaches lies in the large-scale image-text matching pre-training that such models undergo: having been exposed to millions of annotated

image-text pairs, they are expected to acquire strong semantic understanding of visual concepts, including the ability to recognize fire-like objects and to distinguish them from actual fire. FIRE-TASTIC [9] combines a task-specific fire detector with a task-agnostic vision-language model to improve reliability. Fire is first identified frame by frame using a YOLOv8s detector trained to localize flames and smoke separately, producing bounding boxes and confidence scores for potential fire regions. Temporal consistency is enforced by using the detector to mark only those frames where fire candidates persist over time, which also limits how often the more computationally expensive vision-language model is queried. Once a fire candidate is detected with a confidence threshold higher than 0.4, the system asks a pre-trained BLIP-2 model to validate or reject the detector's bounding box. BLIP-2 confirms or rejects fire through keyword spotting in the generated captions: if one of the words chosen at training time (e.g. "burn", "burned", "burning", "burns", "fire", "flame", "flames", "lit", "smoke", "smoking") is present in the caption, the VLM confirms the presence of fire. A fire alarm is raised only when both the detector and the vision-language model agree, substantially reducing false positives while preserving high recall. FOCUS [35] exploits scenario knowledge by tailoring the fire detection pipeline to the typical visual conditions of the monitored environments, accounting for variations in range and background activity, according to the taxonomy defined in [5]. Flames and smoke in individual frames are detected using a YOLOv8s-based two-class object detector, whose head is adapted to distinguish the two phenomena. The training relies on the large carefully curated FOCUS dataset composed of ONFIRE, D-FIRE, and thousands of additional web-sourced images selected to maximize both the diversity of positive samples and the presence of challenging negatives. Temporal consistency is then verified, for long range or high activity scenarios, through rules that track the persistence and stability of detections, monitoring their confidence, position, and size across consecutive frames, to suppress transient false alarms. Finally, for frames that pass this temporal filter in the short range with high activity scenario, with the same approach adopted in FIRE-TASTIC the system queries a BLIP-2 vision-language model, using the detected bounding boxes as spatial prompts, to semantically confirm whether the localized region truly contains fire or smoke. The specific selection of the suited configuration for the operative scenario allows to maximize both recall and precision.

This family of approaches achieves F-score higher than 94%, as shown in Section V, thanks to their ability to effectively manage a wide range of challenging scenarios; the best method obtains an impressive precision higher than 98%, demonstrating the potential of these types of approaches for real fire detection video surveillance systems.

IV. EXPERIMENTAL FRAMEWORK

In the following sections, we outline the experimental framework used to assess the performance of the considered

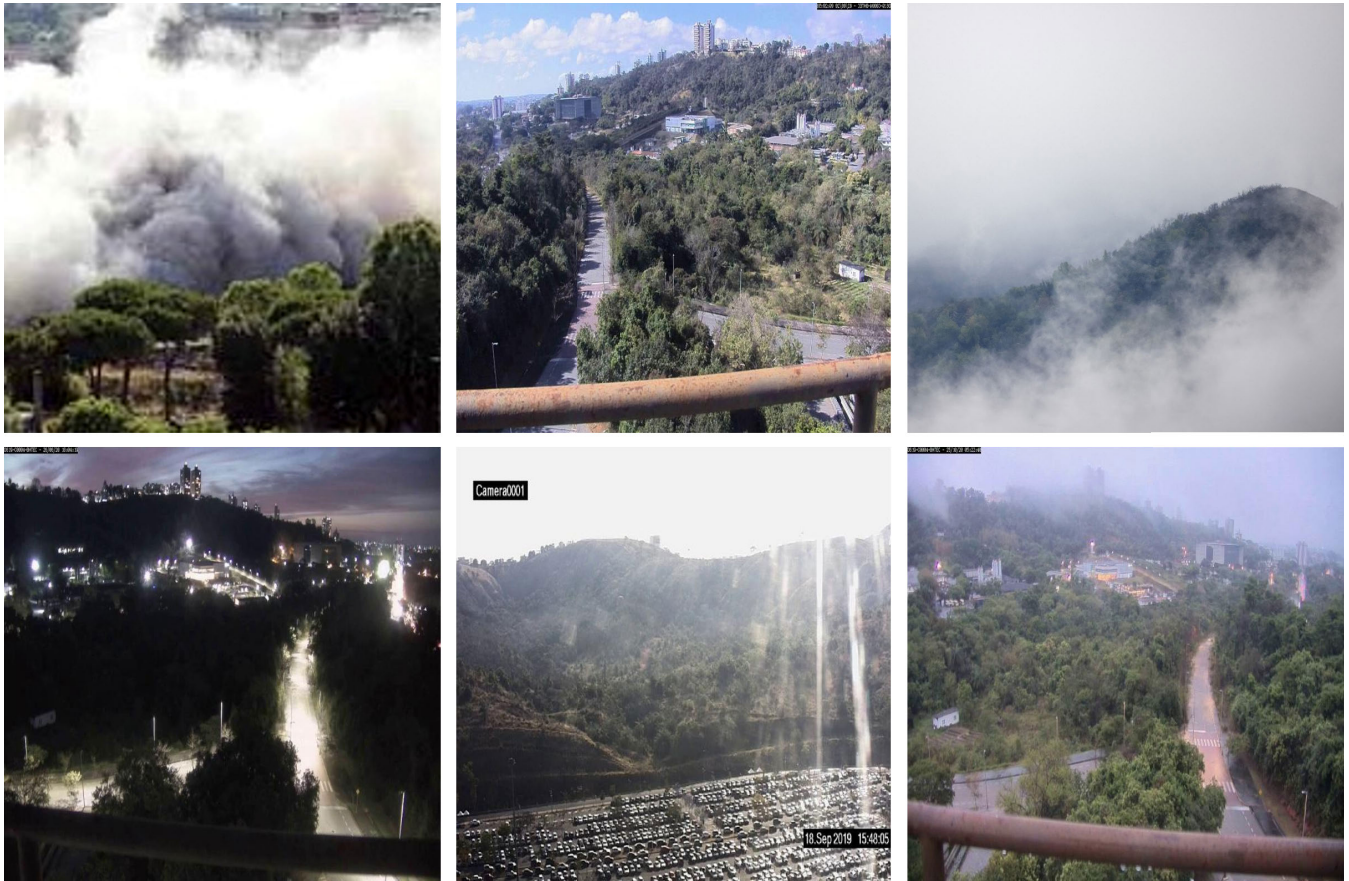


FIGURE 2. Examples of misclassifications produced by methods that do not incorporate semantic confirmation through vision-language models. The first row shows false alarms caused by smoke-like regions erroneously identified as fire, while the second row illustrates cases where intense light sources or flame-shaped objects lead to incorrect flame detections.

fire detection approaches. We first describe the test sets adopted for our evaluation, highlighting their characteristics and the challenges they pose. Then, we detail the metrics adopted to compare the fire detection results.

A. DATASETS

The experimental study relies on two distinct datasets, both designed for benchmarking real-time fire detection in video surveillance.

The first dataset is the collection of clips used in the ONFIRE competitions [7], [8]. It comprises 214 videos, split evenly between 107 fire sequences and 107 non-fire sequences, thereby ensuring a balanced evaluation with respect to both missed detections and false alarms. Each positive video contains a single fire incident, and the precise time instant in which the flames first become visible has been manually annotated. The ONFIRE videos span a broad spectrum of acquisition conditions, with substantial variation in resolution (between 320×180 and 1920×1080), duration (between 10 seconds and 30 minutes), frame rate (between 5 and 30 frames per second), lighting, camera-to-target distance, scene activity, and overall environmental complexity. Such diversity makes the dataset particularly

demanding for automated detection systems. Furthermore, several clips feature elements that may visually resemble fire or smoke, namely headlights, reflective surfaces, moving flags, fog, or cloud formations, posing additional challenges for discriminative algorithms. To enable scenario-dependent performance analysis, every video is assigned to one of the four categories introduced in [5]: low activity short range (LA-SR), low activity long range (LA-LR), high activity short range (HA-SR) and high activity long range (HA-LR). This taxonomy captures differences in scene motion and camera distance, which are known to significantly affect the appearance of flames and smoke. A relevant part of ONFIRE clips fall into high activity settings, where numerous moving objects complicate detection, while the remainder correspond to more controlled low activity contexts. In particular, 60 videos have been collected with low background activity (49 short range and 11 long range), while the remaining 154 depict challenging scenarios with high activity (72 short range and 82 long range).

The second dataset employed in the evaluation is *FIRE-TASTIC* [9], consisting of 156 web-sourced videos, again evenly divided into 78 fire and 78 non-fire samples, annotated with the ignition start time similarly to the ONFIRE

dataset. These sequences exhibit wide variability in resolution (between 320×256 and 1920×1080), duration (between 15 seconds and 15 minutes), frame rate (between 1 and 30 frames per second), illumination, weather conditions, and background motion. Unlike ONFIRE, FIRE-TASTIC annotations do not provide the division in the four scenarios, but the authors report that the dataset includes a large proportion of long range scenes, often characterized by small or partially occluded flame regions. As such, it broadens the range of visual situations and provides a complementary benchmark for assessing generalization capabilities.

Together, the ONFIRE and FIRE-TASTIC datasets offer a comprehensive and challenging test bed, covering diverse real-world conditions from simple near-field scenes to cluttered, dynamic, and long range scenarios.

B. EVALUATION METRICS

The evaluation relies on the metrics defined in the ONFIRE contest [8]: precision (P), recall (R), F-score (F) and Fire Detection Score (FDS). To compute these values, three sets are defined: true positives (TP), false positives (FP), and false negatives (FN).

For each video, the fire ignition time g corresponds to the first frame where the fire becomes visible, as annotated by a human observer. The algorithm detection time p is then compared with this reference. A detection is considered correct (TP) if it occurs no earlier than five seconds before ignition ($p \geq \max(0, g - 5)$). This tolerance should not be necessary because it should be impossible for a method to anticipate the fire notification with respect to the ignition start time. However, the challenge organizers wanted to consider the tolerance value so as not to penalize the methods excessively. The authors of all the considered approaches have reported their results computed according to this protocol. False positives are divided into two types: FP_n , namely the detections in negative videos (normal false positives); FP_e , which are early detections in positive videos (early false positives with $p < \max(0, g - 5)$). False negatives (FN) include all positive videos where no alarm was triggered.

Precision and recall are then defined as:

$$P = \frac{|TP|}{|TP| + |FP_n| + |FP_e|} \quad (1)$$

$$R = \frac{|TP|}{|TP| + |FN| + |FP_e|} \quad (2)$$

where $|TP|$, $|FP_n|$, $|FP_e|$, and $|FN|$ represent the number of true positives, normal false positives, early false positives, and false negatives, respectively. Precision $P \in [0, 1]$ expresses the system's ability to avoid false alarms, while recall $R \in [0, 1]$ measures its capacity to correctly identify real fires.

The F-Score provides a single measure that balances both precision and recall:

$$F = \frac{2 \cdot P \cdot R}{P + R} \quad (3)$$

TABLE 2. Overall performance in terms of precision (P), recall (R), F-Score (F) and Fire Detection Score (FDS) on the ONFIRE test set. The methods are divided according to the taxonomy in the ones exploiting single-image analysis, spatial and temporal analysis and supported by vision-language models. The best value for each metric is highlighted in bold, while the methods are ordered for descending F within their categories.

Method	P	R	F	FDS
Supported by vision-language models				
FOCUS [35]	98.08	95.53	96.68	0.671
FIRE-TASTIC [9]	96.12	92.52	94.29	0.684
Spatial and temporal analysis				
FLAME [41]	93.20	94.12	93.66	0.743
Firebusters [6]	81.15	98.02	88.79	0.731
Slow-Fast [6]	77.48	87.36	82.12	0.549
De Venancio et al. [36]	61.64	88.24	72.58	0.509
MadFire [6]	51.30	88.76	65.02	0.428
Cheng et al. [44]	42.58	87.25	57.23	0.372
Iplab@Unica [43]	41.57	67.65	51.49	0.281
State Key Lab [6]	34.60	96.05	50.87	0.256
AI-KP [6]	45.95	42.50	44.16	0.157
FIRESTART [42]	31.48	53.12	39.53	0.161
KIOS-EVAI [6]	42.55	19.80	27.02	0.047
Single-image analysis				
YOLOv11s [35]	77.02	93.01	84.26	0.659
YOLOv12s [35]	76.61	93.14	84.07	0.658
YOLOv8s [35]	74.42	94.12	83.12	0.647
YOLOv5s [35]	71.93	80.39	75.93	0.430
Tendeiro [7]	40.12	65.69	49.81	0.264
Int. System Lab [33]	28.64	96.72	44.19	0.270
ShuffleNetV2 [34]	69.57	29.91	41.83	0.133
KutnalNet [26]	43.21	32.71	37.23	0.000
FireNet [32]	53.33	14.95	23.36	0.045

Although P , R , and F are widely adopted for evaluating detection accuracy, they do not fully capture practical aspects relevant to real-time fire detection, such as reaction time and computational cost. For this reason, the *Fire Detection Score* (FDS) combines accuracy, timeliness, and efficiency into a single metric:

$$FDS = \frac{P \cdot R \cdot D_n}{(1 + PFR_\delta) \cdot (1 + MEM_\delta)} \quad (4)$$

where D_n is the normalized average detection delay (with respect to a maximum of 60 seconds), PFR_δ is the relative deviation of the processing frame rate (with 10 fps as a reference), and MEM_δ represents the normalized GPU memory usage (with 4 GB as a baseline). The last two terms penalize methods that require excessive computational resources during inference.

V. RESULTS

In the following sections, we present a comprehensive evaluation of all the considered methods. We first report their performance on the ONFIRE test set, both in terms of overall metrics and through a detailed scenario-based analysis that highlights how each approach behaves under different visual and contextual conditions. Then, we provide the results obtained on the FIRE-TASTIC test set, which further validate the trends observed on ONFIRE and offer additional insights into the robustness of the most competitive systems.

A. ONFIRE TEST SET

The analysis of the experimental results on the ONFIRE test set, reported in Table 2, highlights clear performance

differences among the evaluated methods, reflecting the importance of three main factors in video-based fire detection: (i) the quality and the training procedure of the detector; (ii) the presence of explicit temporal analysis; (iii) the ability to distinguish fire and fire-like objects with VLMs.

Among all the considered methods, those that integrate a well-trained detector with temporal analysis and a vision–language model, namely FOCUS and FIRE-TASTIC, achieve the most robust and reliable performance. Their strength lies in their ability to adapt to a broad range of scenarios, including challenging situations encountered frequently in the test set: dynamic backgrounds, fire-like distractors, and long-range scenes with small flame or smoke regions. The achieved FDS values are higher than the methods based only on single-image analysis, showing a positive trade-off between accuracy, notification delay and processing resources. Between the two, FOCUS stands out due to its explicit scenario-based adaptation strategy. By adjusting its inference pipeline depending on the activity level and distance of the observed scene, FOCUS is able to adapt to different application scenarios. This targeted adaptation results in exceptionally high precision, indicating strong resistance to false alarms, while still achieving a remarkably high recall, demonstrating that the system rarely misses true fire events. FIRE-TASTIC, although behind FOCUS in recall and precision, benefits from the complementary strengths of its integrated VLM, which provides semantic disambiguation in challenging cases and contributes significantly to its strong overall performance and to a precision higher than other approaches.

A second tier of methods, including FLAME, FireBusters, and Slow-Fast, relies primarily on a combination of a carefully trained detector and explicit temporal reasoning. Compared to purely detector-based approaches, these methods handle temporal consistency well and are able to suppress short-lived false alarms arising from momentary detection noise or sporadic fire-like patterns. However, their main limitation is the absence of high-level semantic reasoning, which prevents them from robustly distinguishing fire from visually similar distractors, especially in high-activity or long-range scenes. Within this group, FLAME achieves substantially improved precision relative to FireBusters and Slow-Fast, thanks to its careful spatial and temporal filtering based on background subtraction and tracking. These mechanisms effectively suppress many false positives, even in complex backgrounds; moreover, this approach achieves the highest FDS, demonstrating a limited notification delay and processing efficiency. FireBusters maintains the highest recall and the second best FDS, but its detector is highly sensitive to the presence of fire-like patterns; this sensitivity helps to capture early or subtle flame regions, but also increases susceptibility to false alarms. Slow-Fast benefits from a strong spatiotemporal backbone, but its recall–precision trade-off remains behind the more targeted approaches of FLAME and FireBusters, particularly due to challenges in distinguishing fire from rapidly moving

non-fire elements. Overall, these mid-complexity methods demonstrate that temporal reasoning alone can significantly improve performance, but without a mechanism for deeper semantic understanding, they still fall short in the most challenging scenarios.

Below these methods in the leaderboard, the performance of detector-only approaches, such as YOLOv8s, YOLOv11s, and YOLOv12s, reflects the limitations of relying solely on spatial detections without temporal or semantic refinement. These methods achieve high recall, especially due to their training on heterogeneous datasets that expose them to diverse flame textures and shapes, and a good FDS value due to their efficiency. However, their precision remains relatively low, since any fire-like region detected in a single frame may trigger an alarm. Short-lived detection noise, bright objects, or moving lights frequently lead to false positives in the absence of temporal aggregation. Although modern YOLO-based architectures offer excellent speed and image-level accuracy, the analysis reveals that frame-wise predictions alone are insufficient for robust real-time fire detection in dynamic environments.

The remaining approaches in the leaderboard generally suffer from two primary issues: (i) overly complex models (that also reduces the FDS value) relative to the limited dataset size; (ii) single-image analysis approaches trained on small or non-representative datasets. The methods in the first category, typically based on large spatiotemporal architectures or heavy feature fusion, often become overspecialized to the training set. As a result, they occasionally achieve high recall, detecting many true fires, but frequently exhibit low precision, misidentifying moving objects as fire. This behavior stems from the training data bias: in many clips, flame or smoke pixels represent the only motion. Consequently, the models learn an unreliable association between movement and fire presence, leading to systematic errors in high-activity scenarios. As for the approaches in the second category, they lack both temporal analysis and a sufficiently diverse training corpus. As a consequence, they struggle in low precision, due to misclassification of common fire-like distractors, and in low recall, due to limited exposure to varied flame shapes, sizes, and distances. This combination leads to inconsistent performance and makes them unsuitable for deployment in real-world fire detection systems.

B. SCENARIO-BASED ANALYSIS

The scenario-specific evaluation reported in Table 3 further reinforces and makes more explicit the trends observed in the overall ONFIRE results. By analyzing performance across the four categories (LA-SR, LA-LR, HA-LR, HA-SR), it becomes clear how different architectural choices in terms of detector quality, temporal analysis, and the presence of a VLM affect robustness under varying visual and contextual conditions.

Among all the approaches, FOCUS consistently demonstrates superior adaptability. It obtains the highest F-score in every scenario except LA-LR, where it ranks second

TABLE 3. Scenario-based results on the ONFIRE test set in terms of F-Score (F). The methods are divided according to the taxonomy in the ones exploiting single-image analysis, spatial and temporal analysis and supported by vision-language models. The best value for each metric is highlighted in bold, while the methods are ordered for descending F within their categories.

Method	LA-SR	LA-LR	HA-LR	HA-SR	Tot
Supported by vision-language models					
FOCUS [35]	100.00	66.67	96.00	94.74	96.68
FIRE-TASTIC [9]	98.67	66.67	79.37	94.44	94.29
Spatial and temporal analysis					
FLAME [41]	98.67	66.67	96.00	91.23	93.66
FireBusters [6]	93.83	80.00	87.18	80.00	88.79
Slow-Fast [6]	98.70	66.67	72.00	66.67	82.12
MadFire [6]	87.50	44.44	63.16	31.17	65.02
Cheng et al. [44]	84.44	40.00	48.21	46.46	57.23
Iplab@Unica [43]	82.76	66.67	48.89	23.81	51.49
State Key Lab [6]	82.76	30.77	35.00	28.57	50.87
AI-KP [6]	31.11	0.00	29.33	54.24	44.16
KIOS-EVAI [6]	34.04	0.00	35.09	6.90	27.02
Single-image analysis					
YOLOv8s [35]	100.00	50.00	85.37	67.61	83.12
Tendeiro [7]	82.35	66.67	51.69	18.18	49.81
Int. System Lab [33]	78.57	36.36	27.12	16.00	44.19
FIRESTART [42]	63.01	20.00	34.00	24.10	39.53
KutalNet [26]	74.63	0.00	8.82	29.17	37.23
FireNet [32]	30.43	0.00	8.89	31.82	23.36

only to FireBusters. This result aligns with the previously discussed characteristics of both methods: FireBusters exhibits strong sensitivity to faint or distant fire cues, which gives it an advantage in low-activity long-range settings; however, this sensitivity becomes a liability in more dynamic scenarios. In contrast, FOCUS leverages its scenario-based adaptation strategy to select the most suitable inference configuration for each context. In LA-SR, FOCUS benefits from the high sensitivity of its well-trained detector, achieving optimal recall without compromising precision. In long-range scenarios (LA-LR and HA-LR), its integration of spatial-temporal refinement enables a stable trade-off between sensitivity and specificity, effectively suppressing false detections arising from small moving distractors or background dynamics. In the challenging HA-SR setting, the VLM-based confirmation step plays a central role: it provides a substantial precision gain by filtering out complex fire-like objects, leading to the highest performance among all methods.

The results also confirm that fire localization only approaches perform well primarily in LA-SR, where the fire regions are typically larger, closer to the camera, and less prone to confusion with dynamic background elements. In this setting, their strong spatial detectors, trained on heterogeneous datasets, ensure high recall and moderate precision. However, their performance drops markedly in long-range and high-activity scenarios, where single-frame predictions are insufficient to distinguish true fires from sporadic or rapidly moving distractions. The absence of temporal filtering makes these approaches particularly vulnerable to false positives and leads to generally lower F-scores outside the simplest scenario. In contrast, methods that combine a well-trained detector with explicit temporal reasoning, such as FLAME and FireBusters, show clear advantages in the long-range settings. Temporal aggregation allows these

systems to overcome the reduced visibility of flames and the presence of motion artifacts, which are characteristic of HA-LR scenes. FLAME, in particular, demonstrates the benefits of careful spatial-temporal filtering: the integration of background subtraction and tracking substantially improves precision relative to other approaches. FireBusters, despite its stronger susceptibility to false alarms, maintains a high recall, especially when early or subtle flame regions are present. However, both methods struggle in HA-SR, where high-level semantic reasoning becomes essential. In these scenes, only the methods equipped with a VLM, namely FOCUS and FIRE-TASTIC, are able to reliably disambiguate between fire and complex distractors such as headlights, reflections, or rapidly moving non-fire objects.

Finally, the remaining methods in the leaderboard generally achieve satisfactory results only in the LA-SR scenario. Their performance deteriorates significantly in long-range or high-activity contexts, for two main reasons. First, architectures that are overly complex relative to the limited training data tend to overfit specific visual patterns, resulting in unstable generalization across scenarios. Second, simpler image classification or detector only systems based on single-image analysis and trained on small or non-representative datasets lack both the temporal coherence and the semantic richness required to handle ambiguous or cluttered scenes. Consequently, their performance varies widely across scenarios, primarily reflecting the distribution of samples in their training sets rather than true robustness.

It is also worth mentioning that this scenario-based analysis clearly highlights the progressive increase in fire detection difficulty as the scene complexity grows. While most approaches achieve satisfactory performance in low-activity short-range conditions, their robustness degrades in long range and especially in high activity scenarios, where dynamic backgrounds and fire-like distractors become dominant. These results confirm that simple spatial detection or coarse temporal aggregation is no longer sufficient under challenging conditions. Instead, increasingly sophisticated filtering strategies, ranging from motion-aware temporal verification to scenario-adaptive pipelines and semantic confirmation, are required to effectively suppress false alarms while preserving sensitivity. This trend underscores the necessity of coupling accurate fire localization with advanced temporal and semantic reasoning to ensure reliable operation in realistic surveillance environments.

C. FIRE-TASTIC TEST SET

Although the results on the FIRE-TASTIC test set, reported in Table 4, are available only for a subset of methods, specifically FOCUS, FIRE-TASTIC, FireBusters, and YOLOv8s, they further confirm all the previous observations.

Once again, FOCUS achieves the best performance in all the metrics, confirming its strong adaptability to highly variable real-world conditions. FIRE-TASTIC ranks second, trailing FOCUS by a small margin in precision but by a larger margin in recall, highlighting the benefit of scenario-aware

TABLE 4. Comparison in terms of precision (P), recall (R) and F-Score (F) with the methods tested on the FIRE-TASTIC dataset. The methods are divided according to the taxonomy in the ones exploiting single-image analysis, spatial and temporal analysis and supported by vision-language models. The best value for each metric is highlighted in bold, while the methods are ordered for descending F within their categories.

Method	P	R	F
Supported by vision-language models			
FOCUS [35]	98.74	97.43	98.08
FIRE-TASTIC [9]	98.57	94.52	96.50
Spatial and temporal analysis			
Firebusters [6]	85.33	87.67	86.49
Single-image analysis			
YOLOv8s [35]	73.91	93.15	82.42

adaptation in boosting sensitivity to subtle or long-range fire events. FireBusters performs more than ten percentage points below FIRE-TASTIC, reinforcing the idea that in more heterogeneous and unpredictable scenarios, the semantic confirmation provided by a vision–language model becomes even more crucial to maintain robustness and suppress false alarms. YOLOv8s, which lacks temporal reasoning and VLM-based disambiguation, falls an additional four percentage points behind FireBusters. This performance gap further emphasizes the importance of temporal analysis in reducing sensitivity to transient or fire-like distractors, as detector-only approaches remain sensitive to spurious responses when evaluated in diverse and dynamic conditions such as those represented in the FIRE-TASTIC dataset.

VI. DISCUSSION AND POSSIBLE RESEARCH DIRECTIONS

The experimental evaluation across the ONFIRE and FIRE-TASTIC test sets reveals a coherent set of trends that highlight the strengths and weaknesses of current video-based fire detection methodologies. Despite the diversity of architectural paradigms, ranging from detector-only baselines to systems integrating temporal analysis and vision–language models, various aspects consistently emerge as determinants of robust performance and worth of further future investigations.

A. COMBINING SPATIAL DETECTION, TEMPORAL ANALYSIS AND SEMANTIC DISAMBIGUATION

The most reliable systems are those that combine three complementary components: a strong fire detector, explicit temporal modeling, and semantic reasoning through a vision–language model. These methods demonstrate an ability to maintain consistently high precision and recall across a variety of visual conditions, ranging from short-range low-activity scenes to long-range or high-activity scenarios where fire cues become ambiguous or visually suppressed. Their superiority stems from the synergy of three elements: spatial detectors provide sensitivity to flame and smoke patterns, temporal analysis reduces the impact of short-lived false activations, and VLM-based confirmation suppresses semantic ambiguities that arise in the presence of distractors. The results suggest that this triad currently forms the most effective design paradigm for robust and

real-world-ready fire detection. To further advance the state of the art for this type of approaches, future directions may consider the development of lightweight real-time VLM components using PEFT reparameterization, LoRA-like adapters, or prompt-optimization strategies to make semantic confirmation both efficient and deployable on embedded hardware. Moreover, dynamic pipeline adaptation in which the system automatically adjusts detector thresholds, temporal windows, or VLM usage could be inferred at runtime based on scene properties. Finally, joint spatial–temporal–semantic training may enable detectors to learn context-aware representations rather than relying on separate modules with limited cross-component optimization.

B. SCENARIO ADAPTATION

The results clearly indicate that scenario-aware approaches, where models adapt their inference strategy depending on estimated scene characteristics, hold significant promise. Adaptation improves both recall and precision by tailoring the detector sensitivity, temporal filtering, and semantic confirmation to match the task conditions. This principle of adaptive fire detection emerges as one of the most compelling insights from the experiments and represents a promising path forward for the field. Possible future improvements of existing approaches may include the integration of automatic scenario classification modules, capable of identifying scene type, camera dynamics, environmental conditions, and distance categories in real time. Furthermore, a direction could be the definition of unified frameworks where scenario estimation, detection, temporal filtering, and semantic confirmation are tightly integrated into an end-to-end pipeline. As an alternative, meta-learning strategies may be designed to allow the detection model to quickly adapt its internal parameters when encountering unfamiliar scenarios.

C. TEMPORAL ANALYSIS

Methods that perform an effective temporal analysis show a clear advantage over other approaches, particularly in long-range scenes or situations with dynamic backgrounds. Temporal aggregation helps compensate for small or intermittent fire regions, stabilizes predictions, and reduces susceptibility to flickering false alarms. Nonetheless, the absence of semantic reasoning limits the currently available methods in scenarios where distractors closely mimic the appearance or motion patterns of fire. Despite their strong recall and reasonable robustness, these systems tend to plateau in precision when confronted with visually complex or highly dynamic environments. To better deal with these limitations, the design of adaptive temporal filters that modulate their sensitivity depending on scene complexity, motion magnitude, or estimated distance of fire-like regions may help to selectively activate VLM confirmation when temporal filtering indicates uncertainty or ambiguity. In addition, the use of transformer-based or diffusion-inspired temporal backbones could allow to learn long-range temporal dependencies without excessive computational cost.

D. NEED FOR LARGE-SCALE, SCENARIO-BALANCED AND REALISTIC VIDEO DATASETS

A recurring theme across both the test sets is the influence of training data diversity. Models trained on limited or unbalanced datasets exhibit variable performance depending on the similarity between training and testing scenarios. Long-range videos and high-activity scenes remain underrepresented in existing datasets, constraining the generalization ability of many methods. Furthermore, a subset of methods struggles due to excessive architectural complexity relative to the size and diversity of available datasets. These systems often overfit to the specific spatiotemporal patterns seen during training, leading to unstable performance across different scenarios. Conversely, simpler detector-only systems trained on limited datasets lack the breadth of visual experience necessary to generalize to unseen conditions. A recurring theme across both the test sets is the influence of training data diversity. Models trained on limited or unbalanced datasets exhibit variable performance depending on the similarity between training and testing scenarios. To solve these issues, the development of new fire detection datasets with balanced representation across key scenarios (LA-SR, LA-LR, HA-LR, HA-SR), ideally covering hundreds of hours of footage in real environments, could give a notable boost to the performance of fire detection approaches. Moreover, dataset standardization efforts should be done to ensure that future benchmarks include balanced samples across key axes such as activity level, distance, lighting conditions, and distractor types. Similarly, the scientific community could define benchmark protocols that emphasize domain generalization, encouraging methods to perform well under distribution shifts and not just on the scenarios seen during training.

E. CONTINUAL LEARNING AND DOMAIN ADAPTATION

A particularly relevant direction suggested by the observed results, especially with the currently available datasets, concerns the integration of continual learning and domain adaptation frameworks. The experimental analysis strongly indicates that current systems, especially those without semantic reasoning or explicit scenario adaptation, struggle when deployed in environments that differ from the limited distributions present in their training sets. This issue is especially evident in approaches that overfit specific motion or appearance patterns, as well as in detector-only models trained on small or biased datasets. These limitations point toward the need for methods capable of incrementally updating their internal representations as they encounter new visual conditions, camera viewpoints, environmental contexts, or types of fire- and non-fire distractors. Future research may therefore benefit from exploring continual learning strategies specifically designed for video-based fire detection. Such strategies would allow models to incorporate new exemplar videos or scenario types over time, thereby improving robustness without requiring complete retraining from scratch. Crucially, these methods must avoid

catastrophic forgetting, namely the degradation in performance on previously learned scenarios when learning new ones. In parallel, domain adaptation frameworks represent an important complement to continual learning. Given that fire detection systems may be deployed in environments vastly different from those represented in existing datasets (industrial settings, forests, urban environments, etc.), the ability to generalize across domains is essential. Domain adaptation mechanisms could enable the system to align representations from the source domain (training data) with those of the target domain (real-world deployment conditions), even when labeled data in the latter is scarce or unavailable. Finally, a more ambitious direction may be joint continual learning and domain adaptation, enabling systems to progressively acquire new scenario-specific knowledge while remaining resistant to forgetting.

VII. CONCLUSION

In this work, we presented a comprehensive assessment of real-time fire detection methods evaluated on the two most extensive video surveillance benchmarks currently available, namely the ONFIRE and FIRE-TASTIC test sets. By analyzing a broad range of approaches, from lightweight detectors based on single-image analysis to temporally aware architectures and methods integrating vision–language models, we offered a unified perspective on the current state of the art in fire detection and its practical challenges in real-world deployments.

Building on the insights emerging from this comparative study, several promising research directions become evident. A central avenue lies in the combination of spatial detection, temporal analysis and semantic disambiguation. In this field, important is the pursuit of more advanced confirmation mechanisms, particularly those leveraging efficient and adaptable vision–language models enriched through lightweight fine-tuning strategies such as PEFT or prompt-optimization techniques. Moreover, the adaptation to the specific scenario and the definition of more advanced temporal analysis approaches could further improve the performance, especially in long range and high activity scenarios. methods capable of continual learning and domain adaptation, allowing models to evolve with new data while avoiding catastrophic forgetting, represent a crucial step toward long-term reliability in dynamic environments. In addition, the development of more balanced and diverse datasets that accurately reflect the variety of real surveillance conditions, including long-range, high-activity, and fire-like distractor scenarios. Finally, methods capable of continual learning and domain adaptation, allowing models to evolve with new data while avoiding catastrophic forgetting, represent a crucial step toward long-term reliability in dynamic environments.

Through this work, we have carried out not only an evaluation of existing real-time fire detection methods but also a roadmap for future research directions aimed at building more effective, adaptable, and context-aware fire detection

technologies suitable for deployment in increasingly complex real-world scenarios.

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